

Exact Finite Metaplectic Quantization of a Toral Hénon Map and a Finite-Level Aliasing Obstruction

Hénon Route-A Working Series, C128

24 August 2026

Abstract

For the toral Hénon map generated by $A = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}$, we construct an explicit unitary on $\mathbb{C}^7$ and prove exact Weyl Egorov covariance, clock preservation, and antiunitary time reversal. The trace is a finite quadratic action sum whose stationary equation is the Hénon recurrence. Exact arithmetic in $\mathbb{Q}(\zeta_7, \sqrt{7})$ then gives $U^8 = I$ and $\det(I - zU) = (1 - z^8)/(1 + iz)$. Thus the natural finite-level lift is genuine, but its periodic traces cannot carry the exponentially growing real-torus fixed-point counts at all times. This simultaneously closes a natural quantization subgate and proves an aliasing obstruction.

1 Progress over the prior gate

Earlier packages reached, at most, a formal quantum hint: a phase or variational expression without a named Hilbert space and verified operator. Here $\mathcal{H}_7 = \mathbb{C}^7$, the unitary, its observable algebra, its clock, and its reversor are all explicit. Passing that operator gate is the positive advance. The exact finite-order obstruction is equally important: the construction cannot be promoted into a global spectral match by rhetoric.

2 Classical source and reversal

On $\mathbb{T}^2$, let

$$H(x, y) = (3x - y, x), \quad A = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $\det A = 1$ and $RAR = A^{-1}$. Writing $a_n = \text{Tr } A^n$, one has $a_0 = 2$, $a_1 = 3$, and $a_n = 3a_{n-1} - a_{n-2}$. Hence

$$\# \text{Fix}(H^n) = |\det(A^n - I)| = a_n - 2. \quad (1)$$

The primitive-cycle counts follow by Möbius inversion. This real-torus clock is kept separate from the finite quantum clock below.

3 Frozen Weyl convention

Let $N = 7$, $\omega = e^{2\pi i/7}$, and interpret $1/2$ as 4 in $\mathbb{F}_7$. On the basis $|x\rangle$, define

$$\begin{aligned} Q|x\rangle &= \omega^x|x\rangle, & P|x\rangle &= |x+1\rangle, \\ W(q, p) &= \omega^{-qp/2}Q^qP^p, & \mathcal{F}_{xy} &= 7^{-1/2}\omega^{xy}, \end{aligned}$$

$$C_{xx} = \omega^{(3/2)x^2}, \quad U = CF^{-1}.$$

Freezing these signs is essential; changing a Fourier sign changes the classical matrix implemented by conjugation.

The half-phase convention also fixes an odd-level boundary. At the control modulus $N = 8$, the congruence $2h = 1 \pmod{8}$ has no solution, so the literal formula $\omega^{-qp/2}$ cannot be copied unchanged to even level. This does not exclude doubled-phase or other separately defined even-level Weil/metaplectic conventions.

Theorem 1 (exact natural lift). *The matrix U is unitary and, for every $(q, p) \in \mathbb{F}_7^2$,*

$$UW(q, p)U^{-1} = W(3q - p, q). \quad (2)$$

If K denotes coefficientwise complex conjugation and $\Theta = FK$, then

$$\Theta^2 = I, \quad \Theta U \Theta^{-1} = U^{-1}.$$

Proof. The finite Fourier orthogonality sum proves unitarity. Direct use of $PQ = \omega^{-1}QP$, including the half-phase in W , gives (2) first on Q and P and then on all Weyl operators. The same orthogonality sum gives $F\overline{F} = I$ and the two antiunitary identities. The release certificate verifies every matrix entry in $\mathbb{Q}(\zeta_7, \sqrt{7})$, not by a floating tolerance. $\square$

Iteration of (2) implements A^n after exactly n powers of U ; no time rescaling occurs. This is the clock-preservation statement used here.

4 Action sum and stationary recurrence

Since

$$U_{xy} = 7^{-1/2} \omega^{(3/2)x^2 - xy},$$

closing the matrix indices gives, for every $n \geq 1$,

$$\mathrm{Tr} U^n = 7^{-n/2} \sum_{x_0, \dots, x_{n-1} \in \mathbb{F}_7} \omega^{\sum_j ((3/2)x_j^2 - x_j x_{j-1})}. \quad (3)$$

The stationary equations are

$$3x_j - x_{j-1} - x_{j+1} = 0 \pmod{7},$$

which are exactly the coordinate recurrence of H . Equation (3) is an exact finite action sum, not an asserted semiclassical correspondence with the real periodic points in (1).

5 Exact aliasing obstruction

Theorem 2 (finite-level obstruction). *For the frozen convention above,*

$$U^8 = I,$$

and the seven eigenvalues are all eighth roots of unity except $-i$. Consequently

$$\det(I - zU) = \frac{1 - z^8}{1 + iz} \quad (4)$$

$$= 1 - iz - z^2 + iz^3 + z^4 - iz^5 - z^6 + iz^7. \quad (5)$$

Moreover $\mathrm{Tr} U^n = 7$ if $8 \mid n$ and $\mathrm{Tr} U^n = -(-i)^n$ otherwise.

Proof. Cyclotomic matrix multiplication gives $U^8 = I$ and the first eight traces $i, 1, -i, -1, i, 1, -i, 7$. Newton identities reconstruct the characteristic polynomial $(t^8 - 1)/(t + i)$, proving the spectral and determinant statements. $\square$

The classical count in (1) grows like the expanding eigenvalue of A , whereas the fixed-level quantum trace is 8-periodic. Therefore this single finite level cannot carry the classical periodic growth for all n . The mismatch is an exact obstruction, not a failed numerical fit.

6 Evidence and Route-A assessment

The producer uses an exact six-coordinate representation of $\mathbb{Q}(\zeta_7)$ extended by $\sqrt{7}$. It checks 49 unitarity entries, all 49 Weyl observables, 49 entries of each reversal identity, U^8 , and the trace sequence. An independent complex-matrix checker, a SymPy pass, deterministic replay, and thirty-two semantic mutations provide orthogonal controls.

The verified Hilbert space, unitary, exact Egorov identity, clock, and antiunitary justify

(A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION).

This is not A4_ROUTE_B_READY. There is no frozen target validation or divisor census, so A2 and A3 remain failed and Route B remains unauthorized. No Hilbert–Pólya, automorphy, Euler-factor, or root-number claim is made under the active NO_BAD_EULER_OR_ROOT_NUMBER firewall.

7 Conclusion

C128 supplies a fully natural finite metaplectic lift and, at the same time, a theorem explaining why that finite level is not enough. Both statements are source-owned and exact. Any future target-facing step is a separate, explicitly gated experiment.